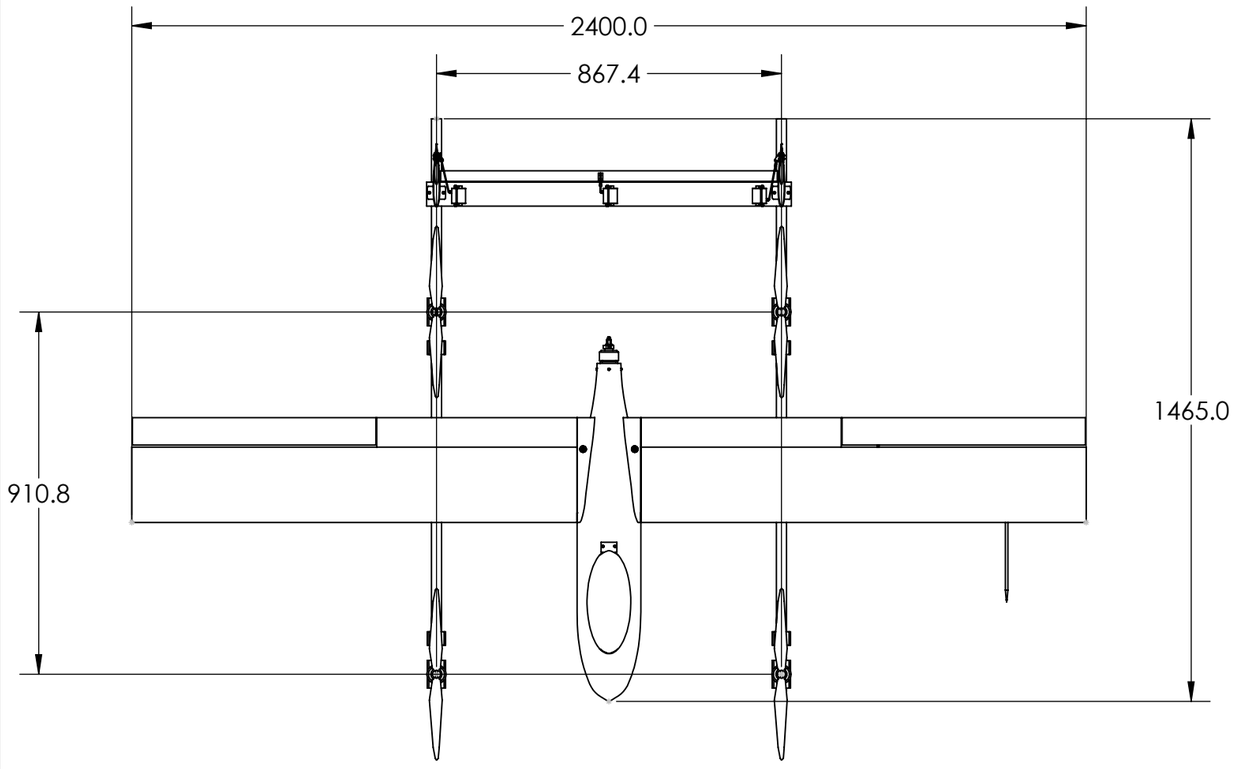
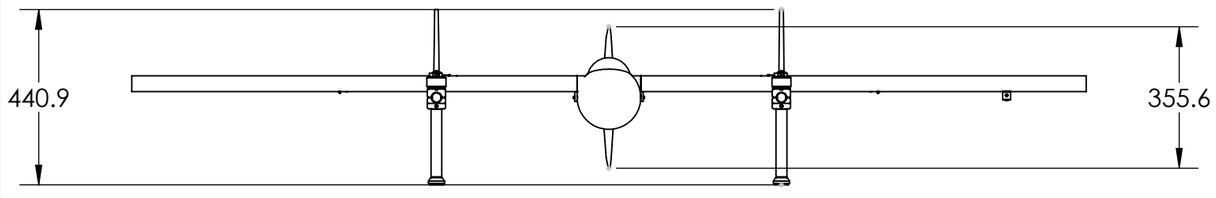
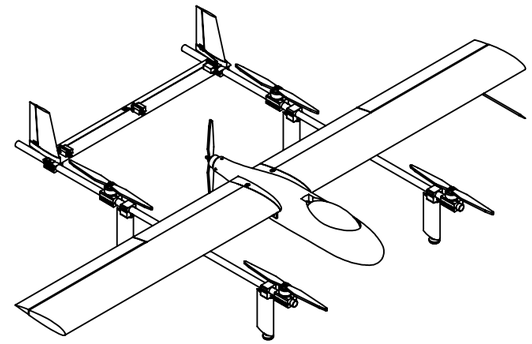


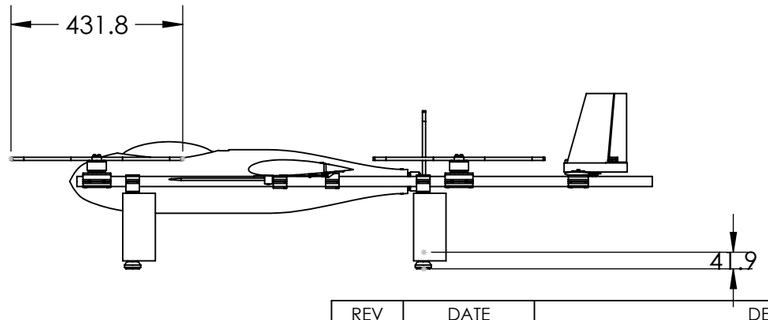
PLAN



FRONT ELEVATION



ISOMETRIC



SIDE ELEVATION

NOTES
1. DIMENSIONS IN MM. VIEWS 1:15, THIRD-ANGLE; ISOMETRIC 1:22.
2. PROPS SHOWN PARKED FORE-AFT (NOTIONAL): THE ESCs CANNOT PARK THEM.
3. SIZED CONCEPT, MTOW FROM analysis/results.json (aero inputs 2026-10-21). EVERY SCREW, NUT,
   WASHER AND INSERT IS MODELLED: SHEETS 2, 3 AND 8.
4. BREAKDOWN FOR TRANSPORT: 2 x M5 WING BOLTS, 4 x M3 PER BOOM CLAMP PAIR, 4 x M3 TAIL SHOE SCREWS; JOINER SLIDES OUT OF THE SADDLE (SHEET 4).

REV	DATE	DESCRIPTION	BY
A	2026-10-18	CONCEPT GEOMETRY FOR DESIGN REVIEW (v0.3.1), 5 SHEETS	DBF
B	2026-10-21	HARDWARE SPEC APPLIED: SPLIT MOUNTS AND CLAMPS, BOOM BUTT SPLICE, WING BOLTS, SHOES, FASTENER MODELLED, TOLERANCES AND FITS, FASTENER SCHEDULE: 9 SHEETS	DBF

	NAME	DATE
DRAWN	DBF	2026-10-21
ENG APPR.		
MFG APPR.		
Q.A.		

COMMENTS:

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN MILLIMETRES. UNLESS NOTED: PRINTED PARTS ±0.5 % OF SIZE, MIN ±0.5 (DESKTOP FDM)
BORES AND INSERT HOLES: SEE FIT TABLE (CLEARANCE)
HOLE POSITION Ø0.4 (MOTOR PATTERN Ø0.2)
CUT TUBES ISO 2768-m: ±0.8 (400–1000), ±1.2 (1000–4000)
ANGULAR ±1°

INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL SEE PARTS LIST
FINISH SEE PARTS LIST
DO NOT SCALE DRAWING

NEXT ASSY | USED ON | APPLICATION

DESIGN-BUILD-FLY

STUDENT CAPSTONE PROJECT, UC BERKELEY.
REV B ISSUED FOR PROTOTYPE BUILD, 2026-10-21.
FIRST-ARTICLE CHECKS M1–M7 OF docs/research/hardware-spec.md APPLY BEFORE ANY PART IS USED. GENERATED FROM analysis/results.json, cad/hardware.json, cad/params.txt.

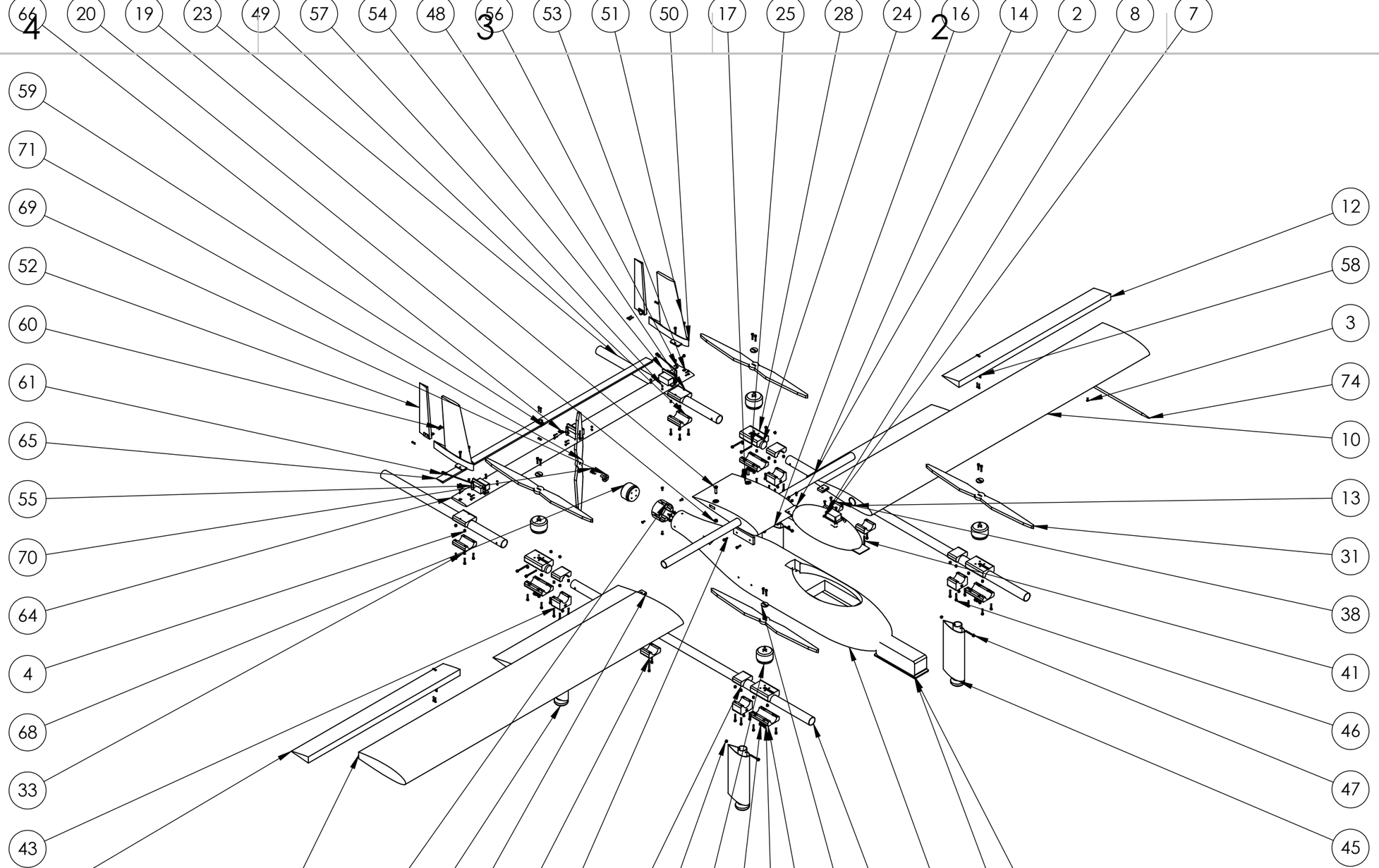
DESIGN-BUILD-FLY

TITLE:

LIFT + CRUISE VTOL GENERAL ARRANGEMENT

SIZE	DWG. NO.	REV
B	DBF-DRW-001	B
SCALE: 1:15	WEIGHT:	SHEET 1 OF 9

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NOTES
1. EXPLODED LAYOUT (VTOL_Exploded.SLDASM): PARTS MOVED BY THE OFFSETS IN cad/params.mis; FASTENERS PULLED 50 mm ALONG THEIR AXES.
2. BALLOON ITEM NUMBERS REFER TO THE PARTS LIST ON SHEET 3. ONE BALLOON PER PART TYPE.
3. JOINT IDs (J01–J20) ARE DEFINED IN research/hardware-spec.md.

EXPLODED ISOMETRIC

DESIGN-BUILD-FLY STUDENT CAPSTONE PROJECT, UC BERKELEY REV B ISSUED FOR PROTOTYPE BUILD, 2026-10-21. FIRST-ARTICLE CHECKS M1–M7 OF docs/research/hardware-spec.md APPLY BEFORE ANY PART IS USED. GENERATED FROM analysis/results.json, cad/hardware.json, cad/params.txt.	NEXT ASSY	USED ON	APPLICATION
UNLESS OTHERWISE SPECIFIED: DIMENSIONS IN MILLIMETRES, UNLESS NOTED PRINTED PARTS ±0.3 % OF SIZE, MIN ±0.5 (DESKTOP FDM) BORES AND INSERT HOLES: SEE FIT TABLE (CLEARANCE/PRESS) HOLE POSITION Ø0.4 (MOTOR PATTERN Ø0.2) CUT TUBES ISO 2768-m ±0.8 (400–1000), ±0.3 (120–400) ANGULAR ±1° INTERPRET GEOMETRIC TOLERANCING PER: MATERIAL SEE PARTS LIST FINISH SEE PARTS LIST DO NOT SCALE DRAWING			

	NAME	DATE
DRAWN	DBF	2026-10-21
CHECKED	—	
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

DESIGN-BUILD-FLY

TITLE: EXPLODED ASSEMBLY WITH BALLOONS

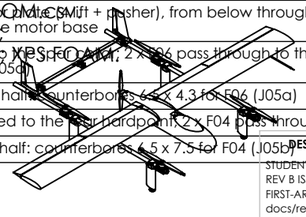
SIZE	DWG. NO.	REV
B	DBF-DRW-001	B
SCALE: 1:12	WEIGHT:	SHEET 2 OF 9

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ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Fuselage_Pod	Fuselage pod 0.85 x 0.16 m, foam + PETG frames + tape (EST): wing-root slot, battery and avionics bays, canopy pocket, tail ring bore 56 x 30 with 4 radial holes, insert holes for the saddle tabs (frame rails in reality)	1
2	Canopy	Avionics hatch blister (EST): foam shell with a 2 mm PETG rear flange (2 x F07 into inserts) and a front tongue under the pod skin (J13)	1
3	F07_M3x8_BHCS	ISO 7380-1 button head M3x0.5 x 8 A2 (18-8) stainless: Light-duty screws into inserts: pusher cup radial screws, canopy hatch, pitot clamp, FC tray (provisional)	11
4	I01_M3_Insert	ruthex RX-M3x5.7 heat-set insert OD 4.6 x 5.7, hole 4 (knurl shown as two bands)	67
5	Battery_Tray	Printed PETG battery tray 165 x 56 x 3 bonded in the battery bay, two 22 x 3 slots for the ShareGoo straps (J18, no fasteners)	1
6	Battery_6S_LiPo	6S 6500 mAh LiPo envelope (dims EST), on the battery tray (J18)	1
7	FC_Tray	Printed PETG flight-controller tray 60 x 44 x 3 with 4 insert bosses on a 45 x 25 pattern (EST: CoreWing F405 Wing V2 hole pattern not confirmed, J15); bonded to the avionics bay floor	1
8	Flight_Controller	CoreWing F405 Wing V2 envelope 52 x 32 x 17 (ArduPilot wiki); hole pattern provisional 45 x 25 (EST)	1
9	Wing_Panel_R	Right wing panel, NACA 2415 (EST), XPS core + tape skin, chord 263 mm; pockets for the spar collar, rear hardpoint, aileron servo tray, root tongue	1
10	Wing_Panel_L	Left wing panel, NACA 2415 (EST), XPS core + tape skin, chord 263 mm; pockets for the spar collar, rear hardpoint, aileron servo tray, root tongue and pitot clamp	1
11	Aileron_R	Right aileron, 26 % chord, 55 % of panel span, tape hinge; pocket for the horn hardpoint	1
12	Aileron_L	Left aileron, 26 % chord, 55 % of panel span, tape hinge; pocket for the horn hardpoint	1
13	Spar_Tube	6061-T6 1.000 x 0.058 in spar socket, 800 mm, bonded in the panel (FT05)	2
14	Wing_Joiner	6061-T6 0.875 x 0.058 in joiner, 760 mm: slip fit in the saddle and sockets (FT01/FT06), captured by the Joiner_Stop discs	1
15	Joiner_Stop	Printed PETG joiner stop 22.4 x 10, bonded in the spar socket 300 mm from the root face: the slip-fit joiner is captured between the two stops once both panels are bolted on	2
16	Wing_Saddle	Wing centre-section saddle: XPS block + printed PETG root plates (EST) carrying the root aerofoil across the pod; joiner bore slip fit (FT06), pillar bores, tongue slots, 6 mm side tabs with 2 x F08 each into pod inserts (J11)	1
17	Bolt_Pillar	Printed PETG wing-bolt pillar dia 16 x 24.1 in the saddle at 70 % c: tongue slot 20.4 x 6.8, 5.7 hole, 10.3 x 1.0 spot-face, M5 nylock pocket 8.2 AF x 8.5 from the lower surface	2
18	Root_Tongue	Printed PETG panel root tongue 20 x 6.4 x 50: 25 mm bonded into the panel root, 25 mm into the saddle pillar slot; 5.7 hole for the M5 x 20 wing bolt	2
19	F09_M5x20_BHCS	ISO 7380-1 button head M5x0.8 x 20 A2 (18-8) stainless: Wing panel retention bolt (one per side, 70 % chord, through the saddle bolt pillar and the panel root tongue)	2
20	N02_M5_Nylock	DIN 985 nylon-insert lock nut M5x0.8, s 8 h 5	2
21	F08_M3x12_SHCS	ISO 4762 M3x0.5 x 12 A2 (18-8) stainless: Wing saddle to pod frame	4
22	Boom_Front	Front boom, 6061-T6 1.000 x 0.035 in, 960.8 mm, cross-drilled 3.4 at 18 mm from the splice end (J04)	2
23	Boom_Rear_Stub	Rear boom stub carrying the tail, 6061-T6, 485.9 mm, cross-drilled 3.4 at 18 mm from the splice end (J04)	2
24	Splice_Plug	Printed PETG splice plug 23.2 x 80 inside the boom butt joint (FT04, 0.02-0.83 mm clearance in the 23.62 bore), cross-drilled 3.4 for 2 x F04	2
25	F04_M3x40_SHCS	ISO 4762 M3x0.5 x 40 A2 (18-8) stainless: Boom splice cross-bolts (rear motor mount) and rear boom-clamp bolts to the wing hardpoint	8
26	W01_M3_Washer	DIN 125 (vendor) M3 flat washer OD 7 ID 3.2 t 0.5	28
27	N01_M3_Nylock	DIN 985 nylon-insert lock nut M3x0.5, s 5.5 h 4	8
28	Motor_Mount_Upper	Printed PA6-CF motor mount, upper half (motor plate + saddle): bore 25.6 (FT02), floor 4.5 under the 4 x M3 on dia 25, counterbores 7.5 (head + washer) from the bore, 4 x RX-M3x5.7 inserts; splice cross-bolt half-grooves (used at the rear station)	4
29	Motor_Mount_Cap	Printed PA6-CF motor mount, lower cap -15.0..-0.6 (1.2 split gap): 4 x 3.4 with 6.5 x 5.6 counterbores for F02; splice cross-bolt half-grooves, counterbore and nut pocket	4
30	Lift_Motor	SunnySky X4110S lift motor (envelope EST; bolt patterns from the maker drawing: base 4 x M3 on dia 25, max depth 3.3; hub 4 x M3 on dia 12, depth 8)	4
31	Lift_Prop	JeeFly 1755 carbon lift prop (planform EST, hub 28 x 8 EST): 4 mm bore + 2 x 3.2 on 12 mm; shown parked fore-aft (notional: BLHeli_S ESCs cannot park props)	4
32	F13_M3x8_SHCS	[illegible] (maker drawing View B-B), supplied with the motor	4
33	F02_M3x16_SHCS	ISO 4762 M3x0.5 x 16 A2 (18-8) stainless: Motor-side bolts (motor mounts, tail clamps), stabiliser/fin shoe screws, and [illegible] lift-prop screws	36
34	F01_M3x8_SHCS	ISO 7380-1 button head M3x0.5 x 8 A2 (18-8) stainless: Rear-boom clamp bolts (lift + pusher), from below through the PA6-CF [illegible] the motor base	20
35	Boom_Clamp_F_Upper	Printed PETG front boom clamp, upper half: counterbores 6.5 x 4.3 for F06 pass through to the collar inserts (J05b)	2
36	Boom_Clamp_F_Lower	Printed PETG front boom clamp, lower half: counterbores 6.5 x 4.3 for F06 (J05a)	2
37	Boom_Clamp_R_Upper	Printed PETG rear boom clamp, upper half: bonded to the rear hardpoint, 2 x F04 pass through (J05b)	2
38	Boom_Clamp_R_Lower	Printed PETG rear boom clamp, lower half: counterbores 6.5 x 7.5 for F04 (J05b)	2

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
39	Spar_Collar	Printed PETG spar collar 32 x 46 bonded around the spar socket at the boom station (front wing hardpoint): 2 x RX-M3x5.7 inserts facing down for the F06 clamp screws	2
40	Rear_Hardpoint	Printed PETG rear wing hardpoint 60 x 46 x 10 bonded flush into the lower surface at 80 % c: 2 x RX-M3x5.7 inserts for the F04 clamp screws	2
41	F06_M3x35_SHCS	ISO 4762 M3x0.5 x 35 A2 (18-8) stainless: Front boom clamp bolts to the spar-collar hardpoint in the wing	4
42	Gear_Clamp_Upper	Printed PETG gear clamp, upper half +0.6..+16.7, 34 x 46, bore 25.6: 4 x RX-M3x5.7 inserts	4
43	Gear_Clamp_Lower	Printed PETG gear clamp, lower half -28.7..-0.6: blind leg socket 25.7 x 14.0 (FT03), 4 x 3.4 with 6.5 x 15.3 counterbores for F03, F05 M3 x 30 cross-bolt 3.4 with 7.5 x 4.0 counterbore and 5.7 AF x 4.3 nut pocket (both clear of the leg tube)	4
44	Landing_Leg	Gear leg: 6061 1.000 x 0.058 in offcut 205 mm + printed dome foot (EST), cross-drilled 3.4 for the F05 retaining bolt	4
45	Gear_Fairing	Printed PETG landing-leg fairing: 3:1 teardrop (NACA 0033 section, chord 81.6) x 170, 0.8 mm vase-mode wall EST, slipped over the leg, TE aft, clocked and taped at the gear clamp (aero drag build-up: faired legs)	4
46	F03_M3x20_SHCS	ISO 4762 M3x0.5 x 20 A2 (18-8) stainless: Gear clamp split screws	16
47	F05_M3x30_SHCS	ISO 4762 M3x0.5 x 30 A2 (18-8) stainless: Landing-leg retaining cross-bolt through gear clamp and leg	4
48	Tail_Clamp_Upper	Printed PETG tail clamp, upper half +0.6..+15.7, 50 x 46, bore 25.6: flat top carries the stabiliser; 4 x RX-M3x5.7 inserts from the split face, 2 from the top for the fin-shoe screws	2
49	Tail_Clamp_Cap	Printed PETG tail clamp, lower cap -15.0..-0.6: 4 x 3.4 with 6.5 x 5.6 counterbores for F02	2
50	Fin_Shoe	Printed PETG fin root shoe: 3 mm floor, 20 mm walls 1.6 thick around the root section (fin bonded in), 32 x 46 flange with 2 x 3.6 for the F02 screws through the stabiliser into the tail clamp (J09)	2
51	Fin	Vertical fin, NACA 0009, lofted taper 65 % tip chord, fixed part 0-68 % chord; full-section root 21 mm sits in the printed shoe; bonded (J09)	2
52	Rudder	Rudder, 30 % of the local fin chord (2 % tape-hinge gap, as the ailerons/elevator), from 22 mm above the fin root to the tip; EMAX ES3005HV via the kit horn (J17)	2
53	Crush_Sleeve	Printed PETG crush sleeve 7 x 3.4 x 8.0 through the stabiliser under the fin-shoe screws (J09)	4
54	Servo_Tray	Printed PETG servo tray (L-bracket): plate 50 x 28 x 3 with the body cutout, 4 x 2.6 holes and M2 nut pockets 4.2 AF x 2.0; 15 mm foot bonded to foam (J14, one part for all 5 servos)	5
55	Servo	EMAX ES3005HV servo 38.6 x 18.8 x 34.9 (J14); mounting tabs, 4 x 2.4 holes at 46 x 10, shaft and 20 mm arm EST (not published, measure on arrival M6)	5
56	F10_M2x10_SHCS	ISO 4762 M2x0.4 x 10 A2 (18-8) stainless: Servo mounting screws (provisional: servo tab hole size and spacing not published)	20
57	N03_M2_Hex_Nut	DIN 934 hex nut M2x0.4, s 4 h 1.6	30
58	Control_Horn	Nylon control horn 11 x 11 x 20 from the 12-set kit (B0B58ZSC8J): 2 x M2 base holes 8 mm apart (EST), 3 pin holes; one per surface (J17)	5
59	Horn_Backplate	Kit horn backplate 20 x 11 x 1.5 (EST) on the far side of the elevator and rudders (J17)	3
60	F11_M2x14_PAN	— M2x0.4 x 14 —: Control-horn screws (supplied with the horn kit, kit item 7)	10
61	Pushrod_Rudder	Du-Bro 230 2-56 pushrod cut to 100 mm (calc from the CAD stations) with nylon Kwik-Link; Z-bend at the servo arm (J17)	2
62	Horn_Hardpoint	Printed PETG aileron horn hardpoint 20 x 12 x 10 bonded flush into the lower surface at 75-83 % c: 2 x 2.6 holes, M2 nut pockets 4.2 AF x 2.0 (nuts captive against the foam)	2
63	Pushrod_Aileron	Du-Bro 230 2-56 pushrod cut to 44 mm (calc from the CAD stations) with nylon Kwik-Link; Z-bend at the servo arm (J17)	2
64	H_Tail	Horizontal stabiliser, NACA 0009 (EST), area from V_h = 0.35; 4 x 7.2 holes for the fin-shoe crush sleeves (J09)	1
65	Elevator	Elevator, 30 % chord, NACA 0009, tape hinge; horn on the top surface at z +40 (J17)	1
66	Pushrod_Elevator	Du-Bro 230 2-56 pushrod cut to 32 mm (calc from the CAD stations) with nylon Kwik-Link; Z-bend at the servo arm (J17)	1
67	Pusher_Cup	Printed PA6-CF pusher motor cup dia 55.6 x 30 (FT11 in the 56 pod ring): 4.5 floor with 4 x 3.4 on dia 25 and relief/wire groove, 4 bosses with radial RX-M3x5.7 inserts for the F07 ring screws	1
68	Pusher_Motor	SunnySky X4110S pusher motor, same part facing aft, carries the M6 bolt-on adapter	1
69	Pusher_Adapter	X4110S bolt-on prop adapter, supplied: M6 x 1.0 thread 14 mm, dia 6 pilot (0/-0.03), flange dia 20 x 5 EST with 4 x 3.2 / 5.7 counterbores on dia 12 (maker drawing View A-A)	1
70	F13_M3x8_SHCS	— M3x0.5 x 8 — (supplied): Pusher M6 bolt-on adapter to the X4110S hub: 4 screws supplied with the motor (maker drawing View A-A, dia 3.2 / dia 5.7 counterbores on dia 12). Length not published: 8 mm EST in the CAD (flange 5.0 EST, counterbore 3.0, engagement 6.0 < 8.0 limit)	4
71	Pusher_Prop_14in	APC 14 x 8.5EP pusher prop (planform EST; hub thickness 8 EST, not published) bolts to the X4110S hub with the supplied M6 [illegible]	1
72	Prop_Washer	[illegible]	1
73	N04_M6_Bullet_Nut	M6 x 1.0 bullet prop nut, supplied with the X4110S (height 19.5 from the maker drawing, captive in the prop)	1
74	Pitot_Tube	Pitot probe under the left wing (Holybro PT40 + MS5525; OD 6 mm EST, length not published)	1
75	Pitot_Clamp	Printed PETG pitot clamp 20 x 20 x 21.5 bonded into the left wing lower surface: 6.2 bore (measured tube OD + 0.2); RX-M3x5.7 insert and F07 pinch screw from below	1

NOTES
1. ITEM NUMBERS ARE THE BALLOONS OF SHEET 2; QTY IS PER AIRCRAFT.
2. PART NUMBER = SOLIDWORKS FILE (cad/models); FASTENER IDs F01–F13, N01–N04, W01, I01 ARE THOSE OF cad/hardware.json AND SHEET 8.
3. PURCHASED ITEMS, PRICES AND VENDORS: docs/purchasing/CO BOM.csv.
4. MATERIALS: 6061-T6 TUBES; A2 STAINLESS FASTENERS; BRASS INSERTS; PETG PRINTS EXCEPT MOTOR MOUNTS AND PUSHER CUP IN PA6-CF; XPS FOAM.



DESIGN-BUILD-FLY STUDENT CAPSTONE PROJECT, UC BERKELEY. REV B ISSUED FOR PROTOTYPE BUILD, 2026-10-21. FIRST-ARTICLE CHECKS M1–M7 OF docs/research/hardware-spec.md APPLY BEFORE ANY PART IS USED. GENERATED FROM analysis/results.json, cad/hardware.json, cad/params.txt.			
NEXT ASSY	USED ON	MATERIAL SEE PARTS LIST FINISH SEE PARTS LIST	
APPLICATION		DO NOT SCALE DRAWING	

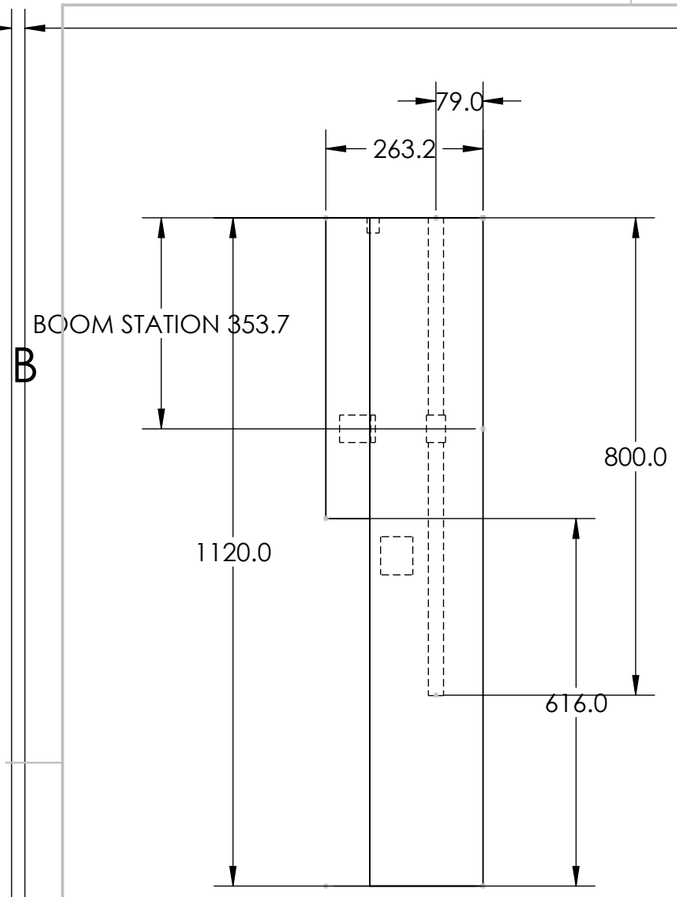
UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN MILLIMETRES. UNLESS NOTED:
PRINTED PARTS ±0.5 % OF SIZE, MIN ±0.5 (DESKTOP FDM)
BORES AND INSERTS PER RX-M3x5.7 / F07 / F01–F13 SPEC
HOLE POSITION Ø0.4 (MOTOR PATTERN Ø0.2)
CUTOUTS ISO 2768-m
ANGULAR ±1°

	NAME	DATE
DRAWN	ER	2026-10-21
CHECKED		
ENG APPR.		
MFG APPR.		
INTERPRET GEOMETRIC TOLERANCING PER:		
COMMENTS:		

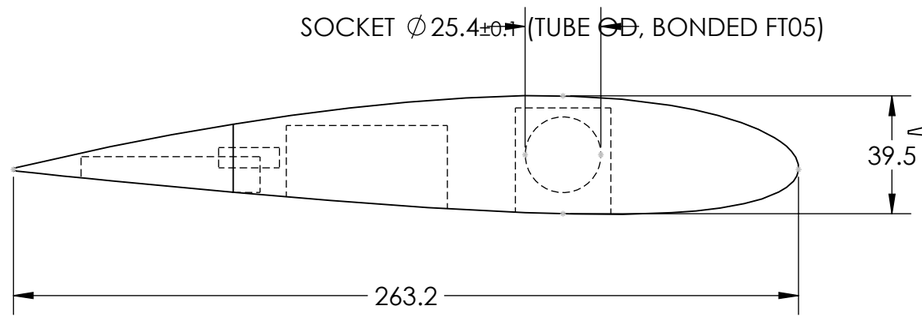
DESIGN-BUILD-FLY
TITLE:
EIFT-CRUISE VTOL
PARTS LIST

SIZE	DWG. NO.	REV
B	DBF-DRW-001	B
SCALE: 1:40	WEIGHT:	SHEET 3 OF 9

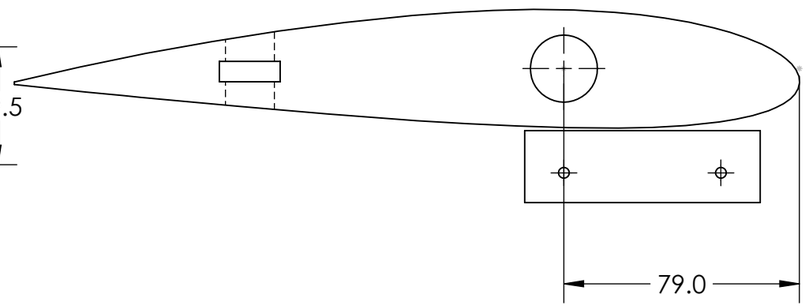
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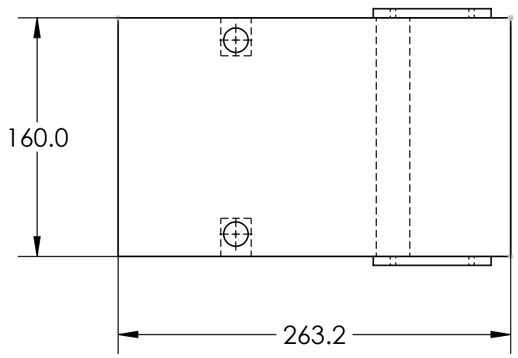
WING PANEL, RIGHT (LEFT IS MIRROR) 1:10



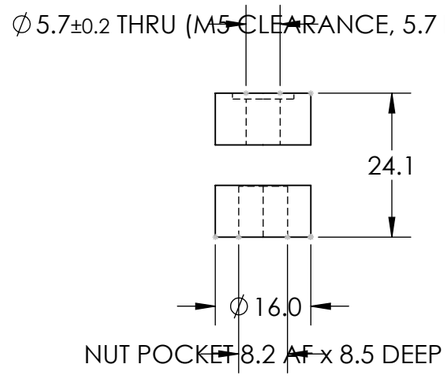
ROOT SECTION 1:2



SADDLE SECTION 1:2



WING SADDLE 1:4



BOLT PILLAR 1:1

NOTES
1. NACA 2415 (t/c FROM SIZING). XPS CORE, PACKING-TAPE SKIN. AILERON 26 % CHORD, 2 % HINGE GAP, OUTER 55 % SPAN.
2. SPAR SOCKET 1.000 x 0.058 in 6061-T6 BONDED 800 mm INTO THE ROOT AT 30 % CHORD (FT05, FOAM-SAFE GLUE).
3. JOINER 0.875 x 0.058 in, 760 mm, SLIP FIT IN THE SOCKETS (FT01: -0.08/+0.53 DIAMETRAL, MEASURE M1) AND IN THE SADDLE (FT06). NOT BONDED: CAPTURED BY TWO PRINTED STOPS 300 mm INSIDE THE SOCKETS ONCE BOTH PANELS ARE ON.
4. WING BOLT J10: F09 ISO 7380 M5 x 20 THROUGH THE PILLAR AND THE PANEL ROOT TONGUE (20 x 6.4 x 50, 25 BONDED) INTO N02 M5 NYLOCK, 1.0 N·m EST. HEAD IN A ∅10.3 x 1.0 SPOT FACE.
5. POCKETS IN THE PANEL (SPAR COLLAR, REAR HARDPOINT, SERVO TRAY, TONGUE, PITOT CLAMP): CUT TO THE PART OUTLINES OF SHEET 9, BONDED WITH FOAM-SAFE GLUE.

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS IN MILLIMETRES. UNLESS NOTED: PRINTED PARTS ±0.5 % OF SIZE, MIN ±0.5 (DESKTOP FDM) BORES AND INSERT HOLES: SEE FIT TABLE (CNC KITCHEN) HOLE POSITION Ø0.4 (MOTOR PATTERN Ø0.2) CUT TUBES ISO 2768-m: ±0.8 (400–1000), ±1.2 (1000–2000) ANGULAR ±1°	DRAWN	DBF	2026-10-21
			CHECKED	—	
			ENG APPR.		
			MFG APPR.		
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.		
		MATERIAL SEE PARTS LIST	COMMENTS:		
NEXT ASSY	USED ON	FINISH SEE PARTS LIST			
APPLICATION		DO NOT SCALE DRAWING			

DESIGN-BUILD-FLY
STUDENT CAPSTONE PROJECT, UC BERKELEY.
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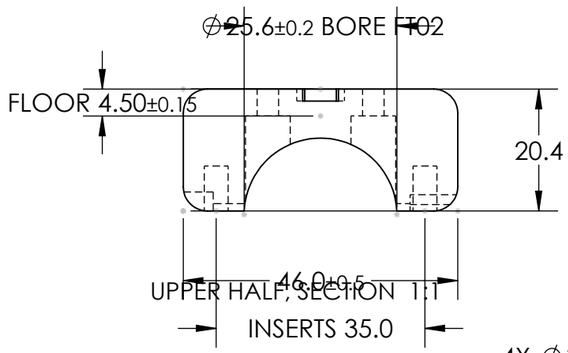
DESIGN-BUILD-FLY

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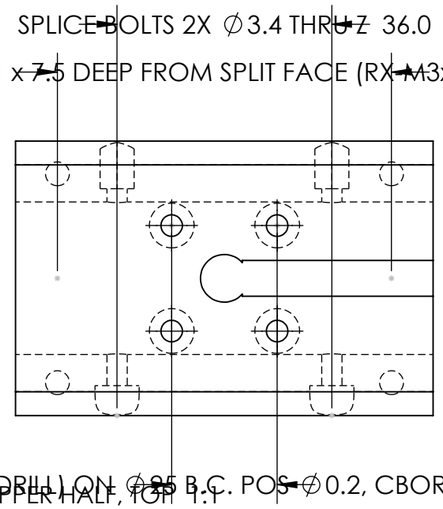
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B	DBF-DRW-001	B

SCALE: 1:10 | WEIGHT: | SHEET 4 OF 9

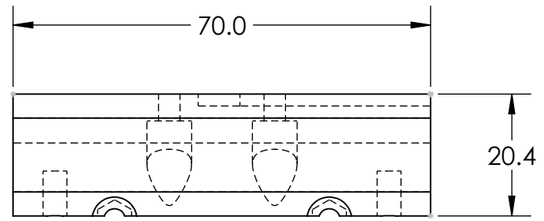
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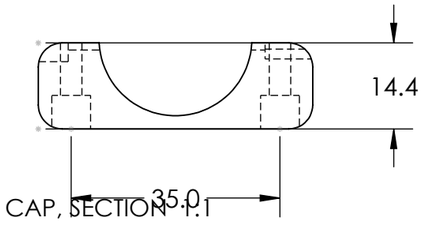
UPPER HALF, SECTION 1:1



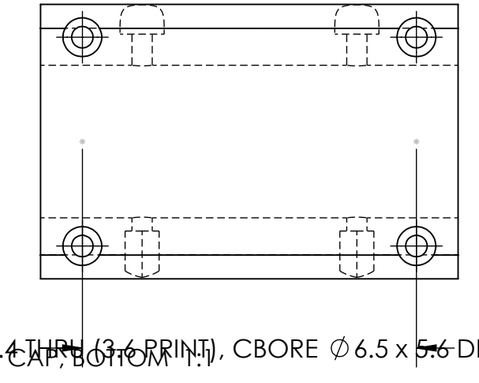
UPPER HALF, TOP 1:1



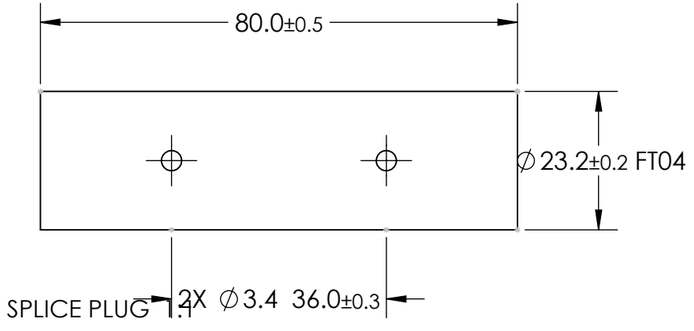
UPPER HALF, SIDE 1:1



CAP, SECTION 1:1



CAP, BOTTOM 1:1



SPLICE PLUG 1:1

NOTES
1. MATERIAL PA6-CF (HDT 173 °C), 4 UPPER HALVES + 4 CAPS. PRINT BORES 0.2 UNDER (COUPON M2), REAM TO 25.6 ±0.2 (FT02). SPLIT GAP 1.2 (0.6 EACH HALF).
2. J01 MOTOR: 4X F01 ISO 4762 M3 x 8 + W01 WASHER FROM INSIDE THE BORE BEFORE THE HALVES CLOSE. ENGAGEMENT 3.0 (LIMIT 3.3, CHECK M4). 0.6 N·m EST, LOCTITE 242 ON THE THREADS ONLY.
3. J02 CLAMP: 4X F02 ISO 4762 M3 x 16 FROM BELOW INTO THE INSERTS, 0.5 N·m EST, NO THREADLOCKER, WITNESS MARK.
4. J04 SPLICE (REAR STATION ONLY): BOOMS BUTT AT THE MOUNT CENTRE OVER THE PLUG; 2X F04 M3 x 40 + W01 + N01 NYLOCK ALONG Z AT x ±18, y 0;
   HEAD CBORE ⌀7.5 x 5.0 (+Z FACE), NUT POCKET 5.7 AF x 8.0 (-Z FACE). DRILL BOTH TUBE WALLS ⌀3.4 THROUGH THE MOUNT AS A JIG, POSITION ±0.3.
5. WIRE GROOVE 6 x 2 ON THE TOP FACE TOWARDS +X; CENTRE RELIEF ⌀8 x 2 FOR THE MOTOR BASE BOSS.

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DESIGN-BUILD-FLY

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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS IN MILLIMETRES. UNLESS NOTED: PRINTED PARTS ±0.5 % OF SIZE, MIN ±0.5 (DESKTOP FDM) BORES AND INSERT HOLES: SEE FIT TABLE (CLEARANCE FIT) HOLE POSITION ⌀0.4 (MOTOR PATTERN ⌀0.2) CUT TUBES ISO 2768-m: ±0.8 (400–1000), ±1.2 (1000–2000) ANGULAR ±1°	DRAWN	DBF	2026-10-21
			CHECKED	—	
			ENG APPR.		
			MFG APPR.		
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.		
		MATERIAL SEE PARTS LIST	COMMENTS:		
NEXT ASSY	USED ON	FINISH SEE PARTS LIST			
APPLICATION		DO NOT SCALE DRAWING			

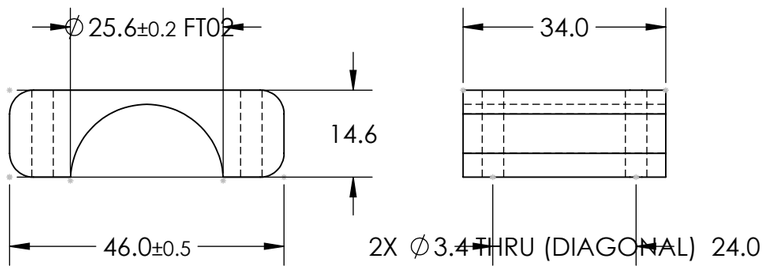
DESIGN-BUILD-FLY

TITLE:
MOTOR MOUNT AND BOOM SPLICE

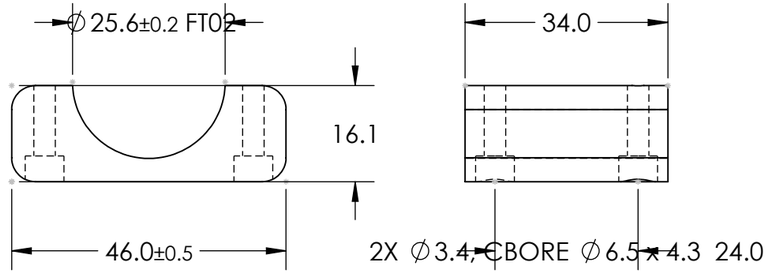
SIZE	DWG. NO.	REV
B	DBF-DRW-001	B

SCALE: 1:1 | WEIGHT: | SHEET 5 OF 9

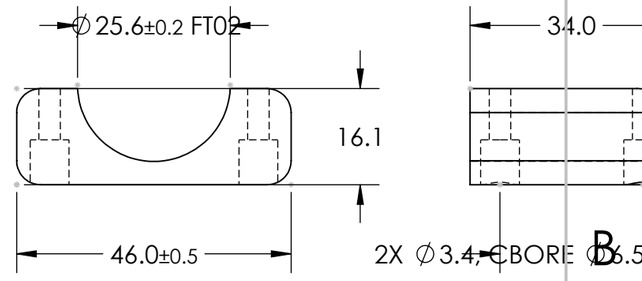
FRONT BOOM CLAMP, UPPER (PETG) QTY 2 1:1



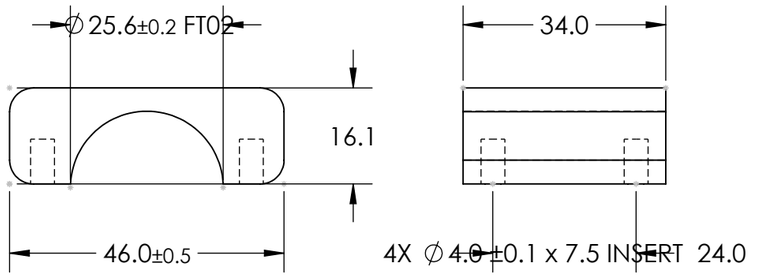
FRONT BOOM CLAMP, LOWER (PETG) QTY 2 1:1



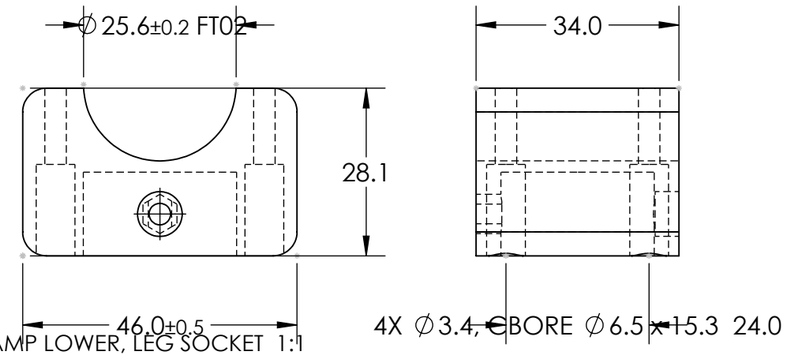
REAR BOOM CLAMP, LOWER (PETG) QTY 2 1:1



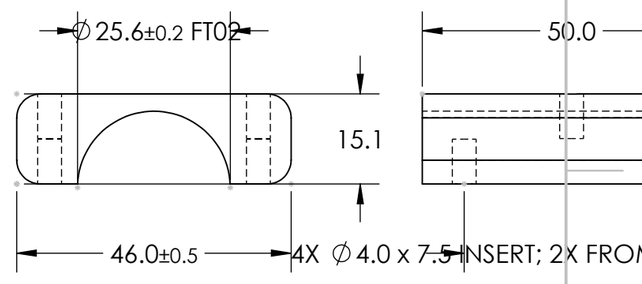
GEAR CLAMP, UPPER (PETG) QTY 4 1:1



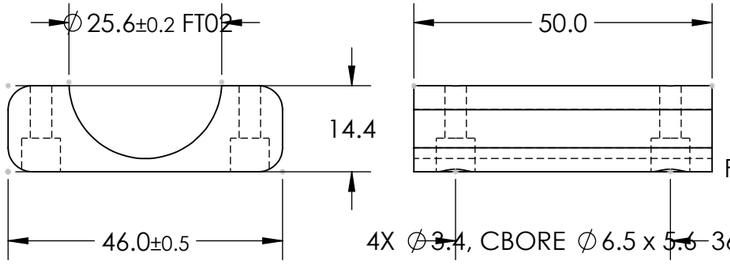
GEAR CLAMP, LOWER (PETG) QTY 4 1:1



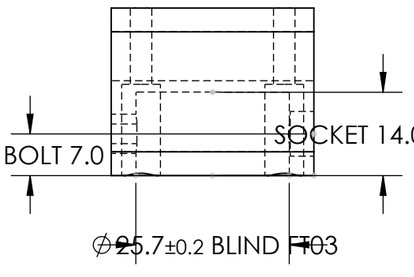
TAIL CLAMP, UPPER (PETG) QTY 2 1:1



TAIL CLAMP, CAP (PETG) QTY 2 1:1



GEAR CLAMP LOWER, LEG SOCKET 1:1



NOTES
1. ALL CLAMPS 46 WIDE (C3), BORE 25.6 ±0.2 (FT02) PRINTED 0.2 UNDER AND REAMED; SPLIT GAP 1.2. REAR BOOM CLAMP UPPER IS THE FRONT UPPER WITH TOP AT 24.3.
2. J05 BOOM CLAMPS: F06 M3 x 35 (FRONT) / F04 M3 x 40 (REAR), 2 PER CLAMP ON A DIAGONAL, THROUGH BOTH HALVES INTO THE WING HARDPOINT INSERTS; e 6.0; 0.5 N·m EST. FIELD BREAKDOWN JOINT.
3. J06 GEAR CLAMP: 4X F03 M3 x 20 INTO THE UPPER-HALF INSERTS, 0.5 N·m EST. J07 LEG: F05 M3 x 30 + W01 + N01 ALONG X AT y -21.7; HEAD CBORE ∅7.5 x 4.0 (+X), NUT POCKET 5.7 AF x 4.3 (-X), BOTH CLEAR OF THE LEG TUBE; LEG DRILLED ∅3.4 THROUGH THE CLAMP.
4. J08 TAIL CLAMP: 4X F02 M3 x 16, 0.5 N·m EST; FLAT TOP CARRIES THE STABILISER; 2 INSERTS FROM THE TOP TAKE THE FIN-SHOE SCREWS (J09).

DESIGN-BUILD-FLY

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NEXT ASSY	USED ON	
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INTERPRET GEOMETRIC TOLERANCING PER:

MATERIAL: SEE PARTS LIST

FINISH: SEE PARTS LIST

	NAME	DATE
DRAWN	DBF	2026-10-21
CHECKED	—	
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

DESIGN-BUILD-FLY

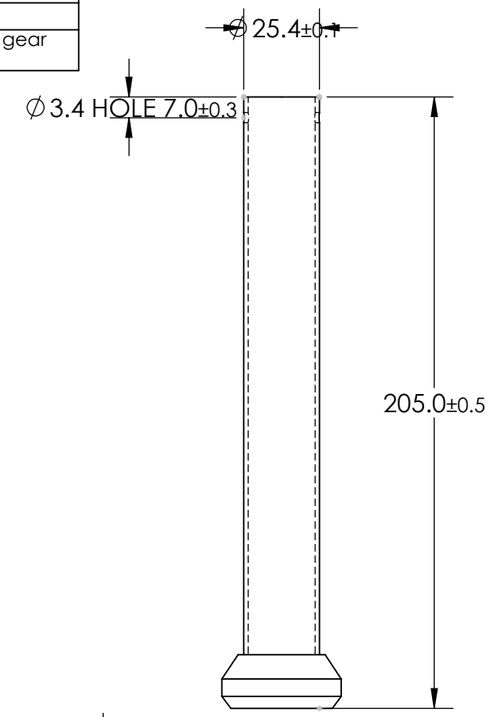
TITLE: BOOM, GEAR AND TAIL CLAMPS

SIZE	DWG. NO.	REV
B	DBF-DRW-001	B

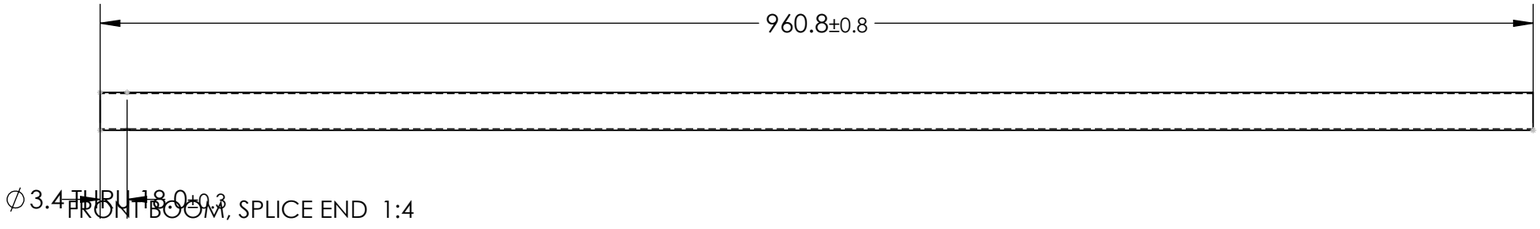
SCALE: 1:1 | WEIGHT: | SHEET 6 OF 9

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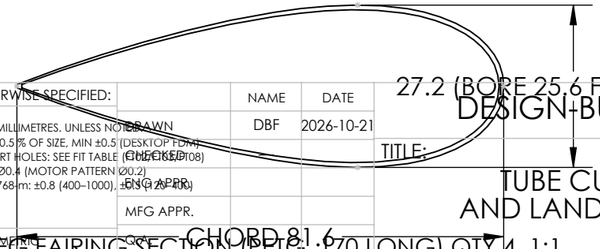
ITEM	PART	STOCK (6061-T6)	CUT LENGTH mm	TOL mm (ISO 2768-m)	QTY	HOLES
T01	Wing spar socket	1.000 x 0.058 in	800.0	±0.8	2	none; bonded in the panel (FT05)
T02	Wing joiner	0.875 x 0.058 in	760.0	±0.8	1	none; slip fit FT01/FT06, lap if tight
T03	Front boom	1.000 x 0.035 in	960.8	±0.8	2	1 x Ø3.4 cross hole (Z) 18 ±0.3 from the splice end, drilled through the mount
T04	Rear boom stub	1.000 x 0.035 in	485.9	±0.8	2	1 x Ø3.4 cross hole (Z) 18 ±0.3 from the splice end
T05	Landing leg	1.000 x 0.058 in	205.0	±0.5	4	1 x Ø3.4 cross hole (X) 7 ±0.3 from the top end, drilled through the gear clamp



LANDING LEG 1:2



FRONT BOOM, SPLICE END 1:4



LEG FAIRING SECTION (PETG, 170 LONG) QTY 4 1:1

NOTES
1. LEGS ARE CUT FROM THE SPAR-SOCKET TUBE OFFCUT: 2 x 800 + 4 x 205 = 2420 mm OF A 96 in (2438 mm) LENGTH. BOOMS: 2 x (960.8 + 485.9) = 2893 mm FROM TWO 72 in LENGTHS.
2. END SQUARENESS 0.5 (EST). DEBURR ALL CUT ENDS AND HOLES. TUBE OD ±0.10, WALL ±0.05 (VENDOR, HS §5a): MEASURE ON RECEIPT (M1).
3. BOOMS BUTT AT THE REAR MOTOR MOUNT CENTRE OVER THE PRINTED SPLICE PLUG (SHEET 5); FRONT AND REAR LENGTHS FROM analysis/results.json geo.segs.
4. LEG FAIRINGS: 3:1 TEARDROP (NACA 0033 SECTION) SLIPPED OVER EACH LEG, TRAILING EDGE AFT, VASE-MODE 0.8 mm WALL (EST), CLOCKED AGAINST THE GEAR CLAMP AND TAPED; NO FASTENERS. DRAG BUILD-UP analysis/aero/aero_results.json: ROUND LEGS 0.0364 OF CD0 -> FAIRED 0.0036.

DESIGN-BUILD-FLY

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INTERPRET GD&T PER ASME Y14.5; TOLERANCING PER ISO 1101.
MATERIAL: SEE PARTS LIST
FINISH: SEE PARTS LIST

	NAME	DATE
DRAWN	DBF	2026-10-21
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

DESIGN+BUILD-FLY

TITLE: TUBE CUT LIST AND LANDING LEG

SIZE	DWG. NO.	REV
B	DBF-DRW-001	B
SCALE: 1:2	WEIGHT:	SHEET 7 OF 9

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ID	STANDARD	SIZE	QTY	ORDER	JOINTS	TORQUE N·m	THREADLOCKER	VENDOR / PART NO.
F01	ISO 4762 (DIN 912)	M3x0.5 x 8	20	30	J01, J12	0.6	Loctite 242, one drop on the las…	Bolt Depot 6379
F02	ISO 4762 (DIN 912)	M3x0.5 x 16	36	50	J02, J08, J09, J03	0.5	none (brass insert in plastic); …	Bolt Depot 6382
F03	ISO 4762 (DIN 912)	M3x0.5 x 20	16	24	J06	0.5	none	Bolt Depot 6383
F04	ISO 4762 (DIN 912)	M3x0.5 x 40	8	12	J04, J05b	0.5	none (nylock)	Bolt Depot 6387
F05	ISO 4762 (DIN 912)	M3x0.5 x 30	4	8	J07	0.5	none (nylock)	Bolt Depot 6385 (M3 x 30 socket cap; number and price not verified, EST)
F06	ISO 4762 (DIN 912)	M3x0.5 x 35	4	8	J05a	0.5	none	Bolt Depot 6386
F07	ISO 7380-1 button head (vendor lists n…	M3x0.5 x 8	11	20	J12, J13, J16, J15	0.4	none	Bolt Depot 7219
F08	ISO 4762 (DIN 912)	M3x0.5 x 12	4	12	J11	0.5	none	Bolt Depot 6381
F09	ISO 7380-1 button head (vendor lists n…	M5x0.8 x 20	2	6	J10	1	none (nylock)	Bolt Depot 6635
F10	ISO 4762 (DIN 912)	M2x0.4 x 10	20	30	J14	0.15	none	Bolt Depot 6365
F11	— (supplier states M2x14 only)	M2x0.4 x 14	10	0	J17	0.15	none (nylon horn)	Amazon (kit item 7) B0B58ZSC8J
F12	ISO 4762 (DIN 912)	M3x0.5 x 16	8	0	J03	0.8	Loctite 242 on the screw tip (me…	Bolt Depot (from F02 or F08 stock) 6382 / 6381
F13	— (supplied; ISO 4762 geometry assumed…	M3x0.5 x 8	4	0	J03b	0.8	Loctite 242 (metal to metal)	SunnySky (in motor box) —
I01	ruthex RX-M3x5.7	M3 x 5.7	67	100	J02 J03b J05 J06 J08 J09 J11 J12 J13 J15 J16	—	—	Amazon (ruthex) B08BCRZZS3
N01	DIN 985 nylon-insert lock nut (vendor:…	M3x0.5 x 4	8	16	J04, J07	—	—	Bolt Depot 4792
N02	DIN 985 nylon-insert lock nut	M5x0.8 x 5	2	6	J10	—	—	Bolt Depot 4794
N03	DIN 934 hex nut	M2x0.4 x 1.6	30	100	J14, J17	—	—	Bolt Depot 7362
N04	M6x1.0 bullet (acorn) prop nut, suppli…	M6x1.0 x 19.5	1	0	J03b	—	—	SunnySky (in motor box) —
W01	DIN 125 (vendor)	M3 7 x 0.5	28	100	J01, J12, J04, J07	—	—	Bolt Depot 4513

NOTES (docs/research/hardware-spec.md §2d, §4)
1. QTY PER AIRCRAFT; ORDER INCLUDES SPARES. F12 PROP SCREWS AND F13 ADAPTER SCREWS: LENGTH PROVISIONAL, CHECK M5 (HUB THICKNESS NOT PUBLISHED).
2. TORQUES MARKED EST IN THE SPEC: CONFIRM INSERT TORQUE-OUT >= 2x ON A COUPON (M3). NO TORQUE DRIVER ON HAND: HANG A KNOWN MASS FROM THE HEX KEY, T = m g L (0.6 N·m = 1.02 kg AT L 60 mm; 0.5 N·m = 0.85 kg; 1.0 N·m = 1.70 kg).
3. LOCTITE 242 ONLY ON METAL-TO-METAL THREADS (F01 MOTOR BASE, F12/F13 HUB SCREWS), ONE DROP ON THE LAST 2–3 THREADS, KEPT OFF THE PA6-CF. NEVER ON NYLOCKS.
4. SCREWS IN BRASS INSERTS GET NO THREADLOCKER: TORQUE, THEN A PAINT WITNESS MARK CHECKED PRE-FLIGHT. REPLACE A NYLOCK THAT TURNS BY HAND.
5. INSERTS RX-M3x5.7 IN Ø4.0 ±0.1 HOLES >= 7.0 DEEP (FT08, INTERFERENCE INTENDED); SOLDERING IRON ~20 °C ABOVE THE PRINT TEMPERATURE, SET PER MATERIAL.
6. CLEARANCE HOLES: ISO 273 NORMAL (M2 2.4, M3 3.4, M5 5.5) PRINTED AT +0.2 AND CHECKED ON THE HOLE COUPON (M2); MOTOR-PATTERN HOLES DRILLED TO 3.4 AFTER PRINTING.
7. MOTOR SCREWS: TIP <= 3.2 mm PROUD OF THE FLOOR BEFORE EVERY MOTOR INSTALL (M4); 3.3 mm REACHES THE WINDINGS.
8. FIELD BREAKDOWN (R4): 2 x F09, 8 x BOOM-CLAMP SCREWS (F06/F04), 4 x F02 SHOE SCREWS; 14 SCREWS IN A LABELLED BAG.

DESIGN-BUILD-FLY

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PRINTED PARTS ±0.5 % OF SIZE, MIN ±0.5 (DESKTOP FDM)
BORES AND INSERT HOLES: SEE FIT TABLE (FT08)
HOLE POSITION Ø0.4 (MOTOR PATTERN Ø0.2)
CUT TUBES ISO 2768-m: ±0.8 (400–1000), ±1.2 (1000–4000)
ANGULAR ±1°

INTERPRET GEOMETRIC TOLERANCING PER:

MATERIAL: SEE PARTS LIST

FINISH: SEE PARTS LIST

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	DBF	2026-10-21
CHECKED	—	
ENG APPR.		
MFG APPR.		
Q.A.		

COMMENTS:

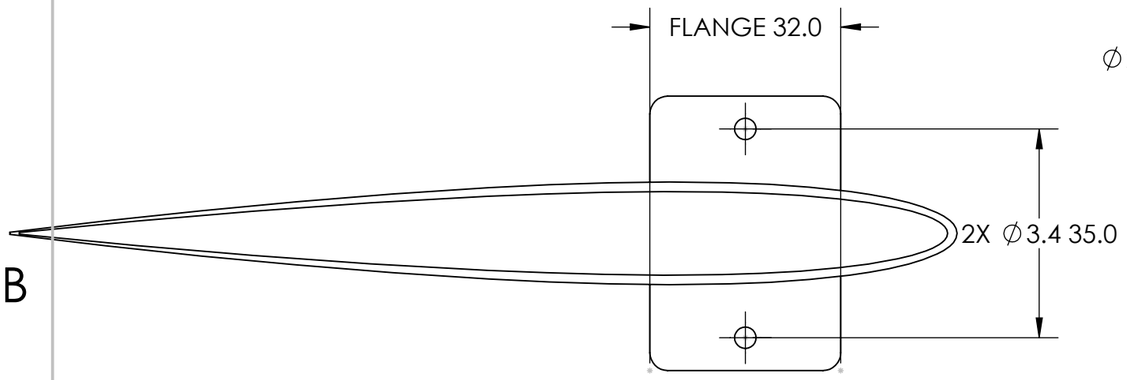
DESIGN-BUILD-FLY

TITLE: FASTENER SCHEDULE AND TORQUES

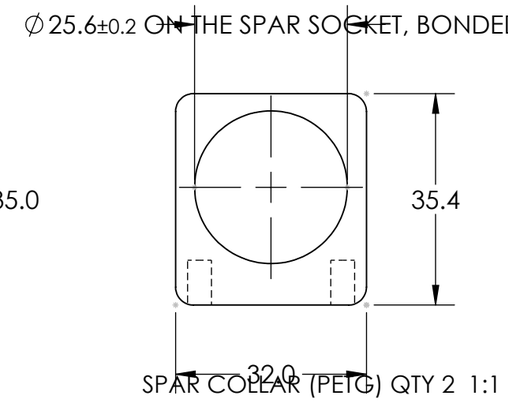
SIZE	DWG. NO.	REV
B	DBF-DRW-001	B

SCALE: 1:1 | WEIGHT: | SHEET 8 OF 9

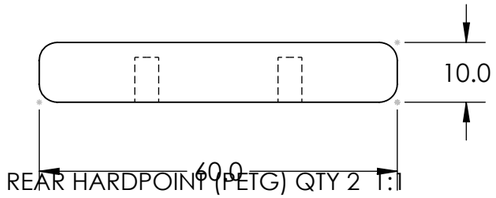
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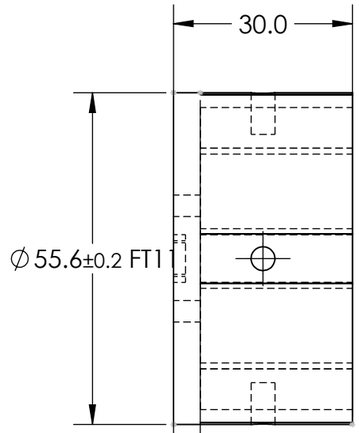
FIN SHOE (PETG) QTY 2 1:1



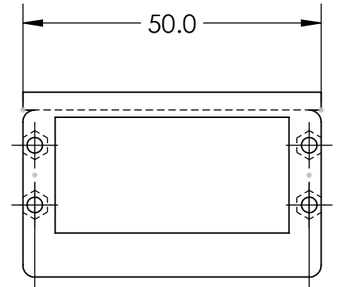
SPAR COLLAR (PETG) QTY 2 1:1



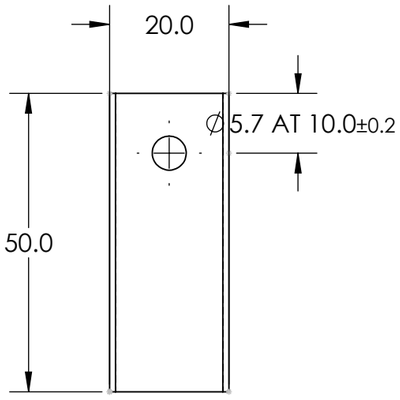
REAR HARDPOINT (PETG) QTY 2 1:1



PUSHER CUP (PA6-CF) QTY 1 1:1



SERVO TRAY (PETG) QTY 1 1:1



ROOT TONGUE (PETG) QTY 2 1:1

NOTES
1. FIN SHOE J09: FLOOR 3.0, WALLS 20 x 1.6 AROUND THE FIN ROOT SECTION (FIN BONDED IN); 2X F02 M3 x 16 THROUGH THE FLANGE, A PRINTED CRUSH SLEEVE 7 x 3.4 x 6.8 IN THE STABILISER AND INTO THE TAIL-CLAMP INSERTS, e 6.2, 0.5 N·m EST.
2. SPAR COLLAR J05a AND REAR HARDPOINT J05b: BONDED FLUSH INTO THE WING LOWER SURFACE AT THE BOOM STATION; INSERTS Ø4.0 x 7.5 FACE DOWN (BOSS WALL >= 1.6).
3. PUSHER CUP J12: 4X F01 + W01 FROM INSIDE INTO THE MOTOR BASE (e 3.0); 4X F07 M3 x 8 RADIAL THROUGH THE POD RING INTO THE BOSS INSERTS, 0.4 N·m EST.
4. SERVO TRAY J14: ONE PART FOR ALL FIVE SERVOS (AILERONS IN THE WING POCKET, ELEVATOR AND RUDDERS ON THE STABILISER TOP): 4X F10 M2 x 10 + N03; TAB HOLES PROVISIONAL (M6).
5. ROOT TONGUE J10: 25 mm BONDED INTO THE PANEL ROOT SLOT, 25 mm INTO THE SADDLE PILLAR SLOT.
6. OTHER PRINTED PARTS (SPLICE PLUG, JOINER STOP, BOLT PILLAR, CRUSH SLEEVE, HORN HARDPOINT, PITOT CLAMP, [illegible]) AS PER THEIR OWN DESCRIPTIONS.

DESIGN-BUILD-FLY
AIRCRAFT DESIGN PROJECT, UC BERKELEY.
REV B ISSUED FOR PROTOTYPE BUILD, 2026-10-21.
FIRST-ARTICLE CHECKS M1–M7 OF docs/research/hardware-spec.md APPLY BEFORE ANY PART IS USED. GENERATED FROM analysis/results.json, cad/hardware.json, cad/params.txt.

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS IN MILLIMETRES. UNLESS NOTED: PRINTED PARTS ±0.5 % OF SIZE, MIN ±0.5 (DESKTOP FDM) BORES AND INSERT HOLES: SEE FIT TABLE (DESKTOP FDM) HOLE POSITION Ø0.4 (MOTOR PATTERN Ø0.2) CUT TUBES ISO 2768-m: ±0.8 (400–1000), ±0.5 (120–400) ANGULAR ±1°	DRAWN	DBF	2026-10-21
			CHECKED	—	
			ENG APPR.		
			MFG APPR.		
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.		
		MATERIAL SEE PARTS LIST	COMMENTS:		
NEXT ASSY	USED ON	FINISH SEE PARTS LIST			
APPLICATION		DO NOT SCALE DRAWING			

DESIGN-BUILD-FLY

TITLE:
PRINTED PARTS
HARDPOINTS AND TRAYS

SIZE	DWG. NO.	REV
B	DBF-DRW-001	B

SCALE: 1:1 | WEIGHT: | SHEET 9 OF 9

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